

Empirical Validation of Score-Derivative SID

Local versus finite targets, matched-model controls, dynamic lag recovery, and robustness

Automated reproducible analysis of the `new_sbtg_neuro` project

14 July 2026

Abstract

This report executes and audits the staged validation plan in *SID-EMPIRICAL-VALIDATION-RUNBOOK.md*. The decisive result is not that one method wins everywhere. Correctly specified normalized laws recover every tested static target and a matched stochastic-gain mixture recovers dynamic distributional lags, but these models are supplied the relevant mechanism family. Among reusable observational estimators, direct moment regression is the most dependable: it recovers mean, covariance, several third-cumulant constructions, and dynamic variance/covariance/third-cumulant lags. Clean score matching sometimes has lower NRMSE, but its effects are often attenuated by roughly one half. Multi-noise and anchored denoising score routes do not provide a general advantage; the anchored route fails dynamic third cumulants and produces a large observation-only false effect. The central bottleneck is the derivative from response score to history tangent and typed readout, not merely predictive-law fit. Results support a targeted predictive-closure program, with score-derived SID retained as a best-case or reusable-law research direction rather than the current default.

Technical summary

Bottom line. Use direct targeted regression for a known functional; use a flexible normalized law when the same fitted law must answer many queries; use a contrast-aligned classifier for a supported finite intervention when its representation can express the relevant geometry. The present score-derived SID implementations are not competitive with those baselines on calibrated typed effects, although clean Hyvärinen plus Hodge is a useful diagnostic and the anchored score improves with very large samples.

- **Validated artifacts.** Tier 1 contains 270 static and 30 dynamic top-level DGP-seed cases, each with a metric table and compressed raw array archive. All 300 archive hashes match, all primary cells contain 30 common seeds, and no fit failed. Tier 2 adds 270 hidden-state/observation-stress cases and 450 static sample/information cases on new seeds 3001–3030. A separately frozen post-hoc fixed-amplitude diagnostic adds 270 cases on paired seeds 4001–4030.
- **Matched models are upper bounds, not evidence of generality.** The static normalized model knows the signed-tilt basis and fits essentially one amplitude. The dynamic “correct MDN” knows the gain vector, component noise, and family, fitting only the mixing law.
- **Derivative gap.** Response-score NRMSE is often 0.04–0.30 while the corresponding reconstructed history tangent is at or above one, especially for quartic, tail, and occupancy mechanisms. Differentiation, centering, and Hodge integration amplify small response-score errors.
- **Dynamic result.** On observed stochastic gain, matched mixture NRMSE is about 0.05 and direct moment regression about 0.22–0.26 across variance, covariance, and third cumulant. The anchored score reaches variance NRMSE about 0.50 but is slightly under-calibrated, and third-cumulant NRMSE exceeds 1.3 with a wrong-sign slope.
- **Robustness result.** With a hidden modulator, direct and Gaussian log-variance recovery improve smoothly with sample size. The anchored score is unstable at 8,000 samples and becomes usable

only at 32,000, still behind direct baselines. Under an observation-only null its false variance effect is comparable to the true active clean effect.

- **Status of the broad claim.** The evidence is a rigorous diagnostic validation of these implementations and DGP families, not a validated claim that SID recovers arbitrary moments in unknown high-dimensional nonlinear systems. Dense unknown response subspaces, multiple gain factors, and general neural predictive backends remain unresolved.

1 Question and estimands

The scientific question is whether a predictive-law model can reveal how a source history direction changes a chosen future-response functional, particularly when the conditional mean is invariant but variance, covariance, skewness, tails, or mode occupancy change. Two targets must remain separate.

1.1 Local target

For conditional law P_h and history direction v ,

$$\ell_v(y; h) = D_v \log p(y | h), \quad D_v T(P_h) = E_{P_h}[\text{IF}_T(Y; P_h) \ell_v(Y; h)].$$

The response score $s_y = \nabla_y \log p(y | h)$ is not ℓ_v . A score route first estimates s_y , differentiates it with respect to history to obtain a mixed field, reconstructs a centered history tangent (when topology permits), and only then integrates it against an influence function.

1.2 Finite target

For a supported central contrast,

$$\Delta_{T,\delta}(h) = \frac{T(P_{h+\delta v}) - T(P_{h-\delta v})}{2\delta}.$$

A balanced endpoint classifier estimates a density-ratio witness for this finite signed measure. This is not interchangeable with the local derivative, and the leaderboards in this report never mix them.

1.3 Recovery and advantage rules

The preregistered numerical recovery rule requires an upper 95% DGP-bootstrap bound below $\text{NRMSE} = 1$, a calibration slope in $[0.8, 1.2]$, correct signs, and an acceptable nuisance false-call rate. A superiority claim additionally requires the upper paired error-ratio bound below 0.90 under the same information access. Because the Tier 1 target-only cases do not emit every nuisance functional, this report distinguishes *numerical recovery* from the stronger fully validated recovery label.

2 Methods map

Table 1: The four layers that earlier reports sometimes conflated.

Layer	Examples	Object learned or transformed	Primary mode	failure
Predictive back-end	raw-moment ridge, Gaussian likelihood, matched tilt, matched gain mixture	$P(Y \mid H)$, selected moments, or structural parameters	misspecification; target variance	
Score estimator	clean Hyvärinen, multi-noise DSM, Gaussian DSM, anchored residual energy	$\nabla_y \log p_\sigma(y \mid h)$	smoothing bias; derivative instability	
SID adapter	density AD, ratio critic, energy centering, score secant, Hodge	$D_v \log p$ or a finite density-ratio witness	centering; support topology; access mismatch	
Readout	mean, covariance, cumulants, tail, occupancy, lag tensor	typed $D_v T$ or $\Delta_{T,\delta}$	influence-function variance; basis mismatch	

2.1 Methods tested in the new confirmatory suite

Direct raw moments (B2).

Polynomial history basis with heads for raw moments through order four (and event indicators where required), analytically converted to central functionals and differentiated.

Matched normalized law (S1/A1).

For M3–M5, the true signed-tilt basis $\psi(y)$ is supplied and only its amplitude is fitted. For mean and covariance, Gaussian moment regressions are used. This is a specification upper bound.

Ratio critic (A2).

Logistic joint-versus-shuffled classifier with explicit history–response interactions; the history derivative of its logit is centered and integrated against a typed influence function.

Clean Hyvärinen plus Hodge (S2/A7).

Polynomial response-score model fit by clean score matching; its history derivative is integrated over response space and centered.

Multi-noise DSM plus Hodge (S5/A7).

Polynomial denoising score fit at $\sigma \in \{0.03, 0.06, 0.12, 0.25\}$.

Gaussian DSM control (S6).

Conditional Gaussian score obtained from fitted low-order moments; this diagnoses how much is available from a Gaussian core.

Anchored residual-energy DSM (S7).

Multi-noise residual score around the Gaussian core, with development-only ridge and noise ladder frozen before confirmation.

Finite adapters.

Direct endpoint Monte Carlo, finite differences of moment regression, normalized-law endpoints, endpoint classifier, signed Riesz dictionary, and a score-integrated endpoint density.

3 Reproduction audit

The frozen legacy core contains 784 completed cases and the finite panel contains 212,832 valid metric rows. A new nine-case subset crossing G1, G3, and G8 with Gaussian likelihood, EDM diffusion, and ratio critic was rerun. All 117 scientific metric values matched bitwise; 18 runtime-efficiency values varied

with execution. The historical source-tree hash and current source-tree hash differ, so this is reported as a compatibility reproduction rather than exact source reproduction.

The G8 contrast-aligned developmental experiment remains important: across four independent generator parameterizations (eight generator/data cells), a motif MLP has median typed-effect relative error 1.13%, a typed quartic classifier 1.16%, and a full-path/history MLP 3.05%. Linear and quadratic negative controls recover essentially none of the effect. Thus the G8 failure of generic generative readouts is not non-identifiability; it is representation/readout misalignment.

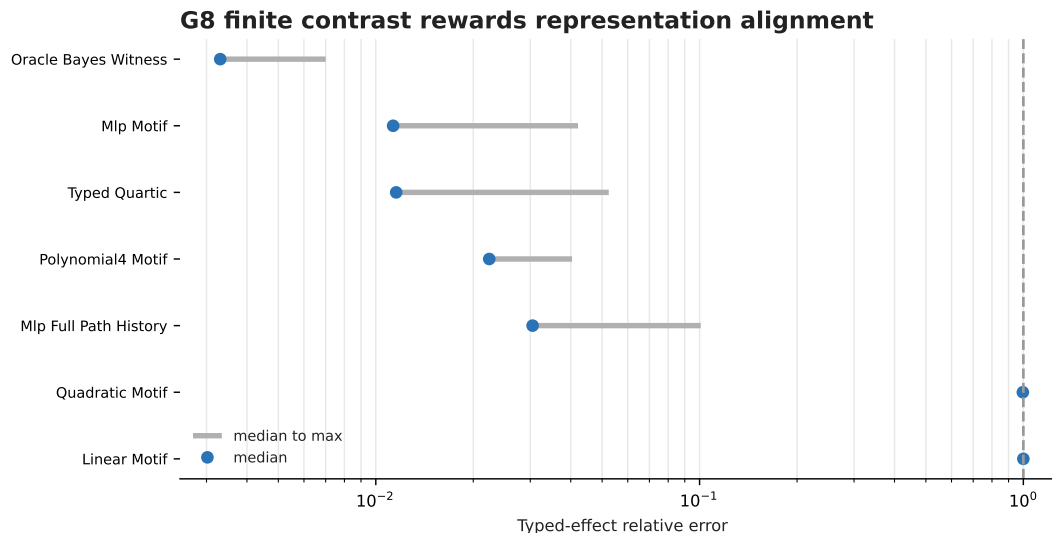


Figure 1: G8 typed finite contrast: aligned classifiers recover the quartic path effect while lower-order negative controls do not. This is developmental evidence with four independent systems, not a 30-system confirmation.

4 Oracle gates

Stage 0 executed 53 algebra/software checks covering normalization, positivity, moment orthogonality, local and finite identities, Gaussian corruption, dynamic lag derivatives, classifier conversion, and Hodge reconstruction. All passed. The topology ladder is decisive: Hodge reconstruction is numerically exact while bridge mass is positive, but a pure disconnected mixture-weight change is globally unrecoverable from within-component response gradients.

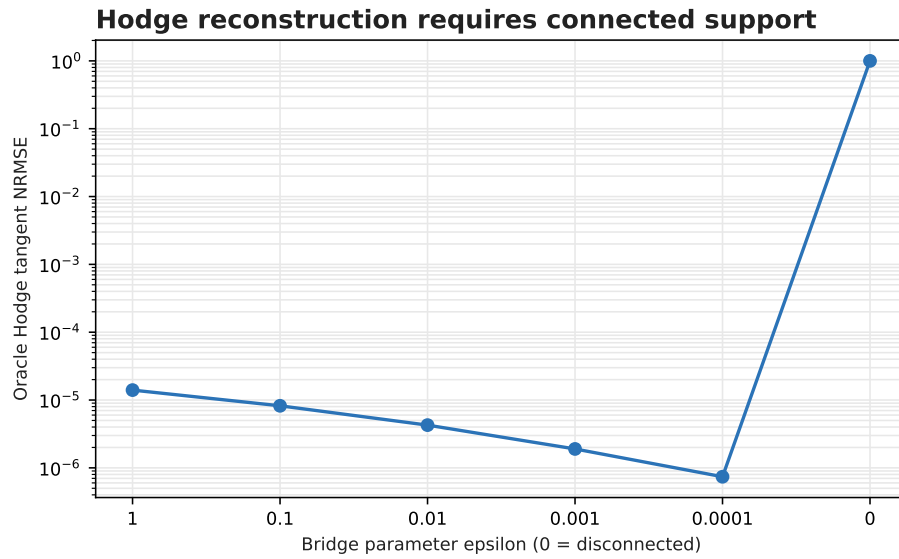


Figure 2: Oracle Hodge reconstruction across the registered support-topology ladder. The disconnected endpoint is an identifiability failure, not an optimizer failure.

The oracle gates establish that the requested typed quantities are mathematically identifiable in the connected static systems and that the numerical readout formulas are correct. They do not establish that a learned score supplies an accurate mixed field.

5 Score-estimator results

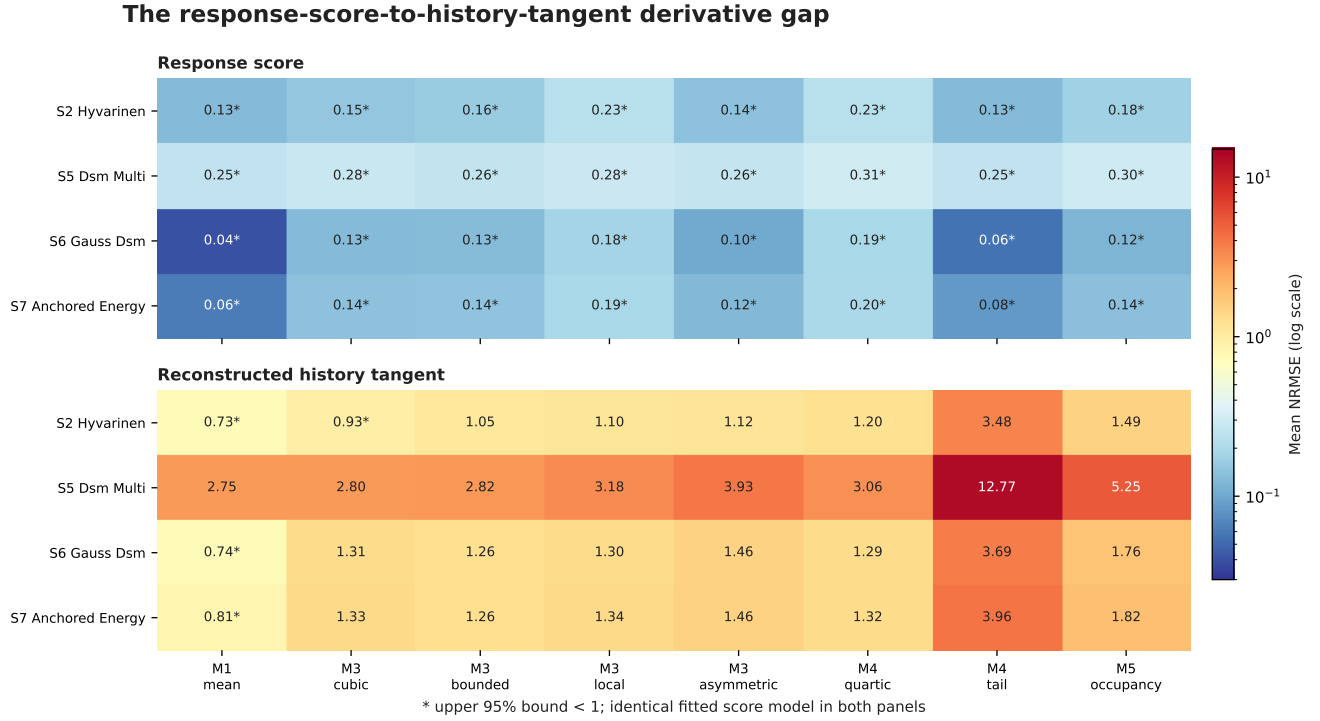


Figure 3: Response-score and reconstructed history-tangent NRMSE use the same fitted score models. Accurate response scores frequently fail after history differentiation and Hodge reconstruction. A star denotes an upper 95% bound below one.

The clean Hyvärinen estimator is the strongest generic score route in the static suite. On the M3 variants its typed NRMSE is roughly 0.54–0.57, but calibration slopes are only about 0.47–0.49. The apparent NRMSE win over direct moments is therefore shrinkage, not calibrated recovery. Multi-noise DSM is consistently worse, and the Gaussian and anchored controls remain near zero-effect behavior on M3 while exploding on fourth-order targets.

The derivative gap supports the following failure chain:

small score error $\nRightarrow$ small mixed-field error $\nRightarrow$ small centered-tangent error $\nRightarrow$ calibrated typed effect.

Stabilization reduces numerical variance but imposes a strong low-order/zero-effect bias. More denoising noise does not repair this structural mismatch.

6 Local SID leaderboard

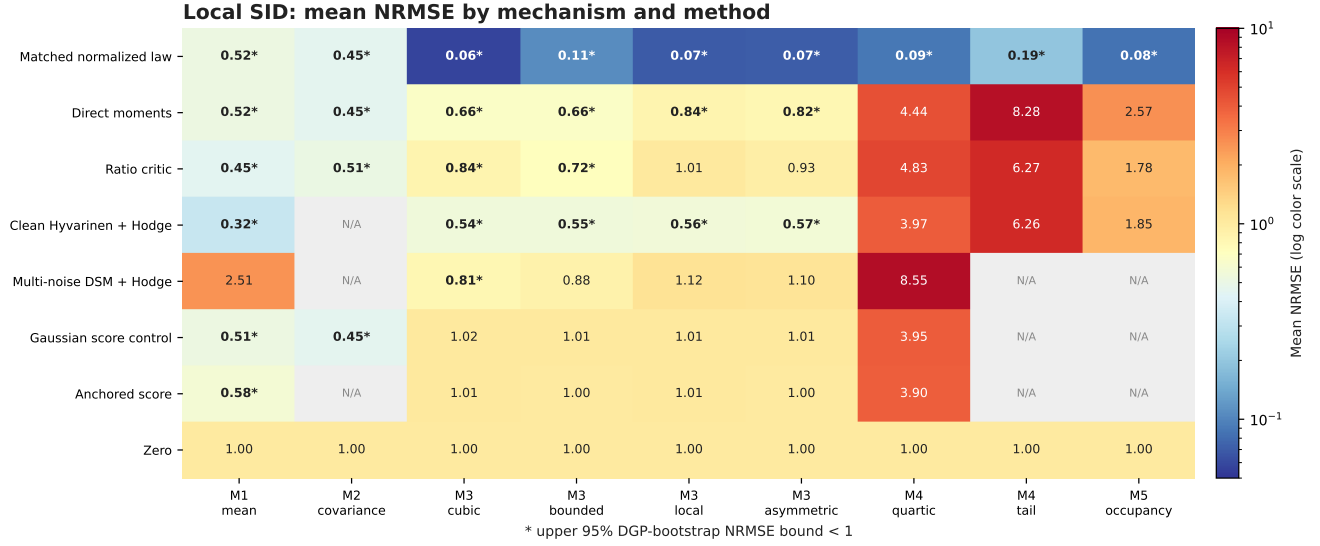


Figure 4: Mean local NRMSE over 30 top-level DGP seeds. The matched normalized law is separated visually because it is supplied the DGP tilt basis. Stars denote an upper 95% bound below one; they do not by themselves certify calibration or nuisance specificity.

Table 2: Selected local results. Values are mean [95% DGP-bootstrap interval]. Partial means NRMSE recovery without registered calibration.

Target	Method	NRMSE	Calibration slope	Calibrated?
M1 mean	Direct moments	0.515 [0.421, 0.622]	1.010 [0.955, 1.063]	Yes
M1 mean	Ratio critic	0.447 [0.391, 0.511]	1.008 [0.931, 1.082]	Yes
M1 mean	Clean Hyvärinen + Hodge	0.324 [0.290, 0.361]	0.836 [0.794, 0.875]	Yes
M2 covariance	Direct moments	0.453 [0.383, 0.526]	1.021 [0.968, 1.073]	Yes
M2 covariance	Ratio critic	0.506 [0.446, 0.570]	1.050 [0.954, 1.151]	Yes
M2 covariance	Gaussian score control	0.453 [0.385, 0.527]	1.021 [0.968, 1.071]	Yes
M3 bounded skew	Matched normalized law	0.105 [0.080, 0.133]	1.008 [0.961, 1.055]	Yes
M3 bounded skew	Direct moments	0.658 [0.562, 0.763]	0.990 [0.916, 1.063]	Yes
M3 bounded skew	Ratio critic	0.718 [0.620, 0.825]	1.003 [0.888, 1.126]	Yes
M3 bounded skew	Clean Hyvärinen + Hodge	0.547 [0.515, 0.579]	0.489 [0.450, 0.530]	Partial
M4 quartic	Matched normalized law	0.088 [0.069, 0.110]	0.993 [0.956, 1.031]	Yes
M4 quartic	Direct moments	4.443 [3.900, 5.041]	0.684 [0.194, 1.182]	No
M4 quartic	Clean Hyvärinen + Hodge	3.970 [3.697, 4.249]	0.099 [-0.223, 0.419]	No
M4 tail	Matched normalized law	0.193 [0.099, 0.296]	0.807 [0.713, 0.895]	Yes
M4 tail	Direct moments	8.278 [6.962, 9.735]	1.025 [0.071, 1.947]	No
M5 occupancy	Matched normalized law	0.084 [0.051, 0.119]	0.916 [0.882, 0.950]	Yes

Target	Method	NRMSE	Calibration slope	Calibrated?
M5 occupancy	Direct moments	2.569 [2.071, 3.112]	0.897 [0.636, 1.172]	No

The general observational conclusion is heterogeneous. Direct moments recover M1 mean, M2 covariance, and all four M3 third-cumulant constructions numerically, with near-unit slopes. The ratio critic is competitive on M1/M2 and two M3 variants but needs conditional evaluation draws in this implementation. No general method recovers M4 quartic at medium information. M4 tail and M5 occupancy are additionally information-bound because positivity caps achieved Λ at 0.128 and 0.949, respectively. Their matched-model success proves the targets exist in the data but does not remove the information limitation for a generic learner.

7 Finite leaderboard

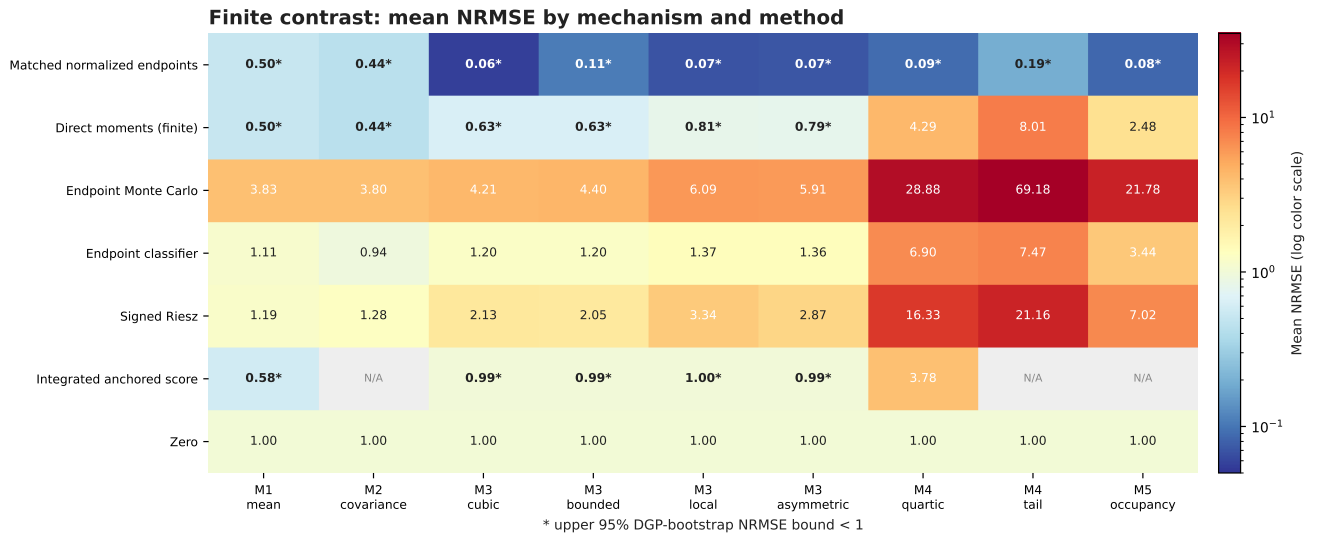


Figure 5: Finite-target NRMSE at $\delta = 0.12$. Local and finite values are never pooled. Endpoint labels or queries are stronger information access than observational regression.

Finite differences of fitted moment regression behave similarly to the local direct method. Naive independent endpoint Monte Carlo is extremely noisy because two noisy functional estimates are divided by 2δ ; common random numbers or analytic model moments are necessary. The generic endpoint classifier and signed Riesz dictionary do not recover most static channels at this sample size. This does not contradict G8: the successful G8 classifier uses a contrast-aligned quartic/motif representation and far more evaluation draws.

8 Higher-order findings

The experiments distinguish four ideas that should not be called simply “higher-order recovery.”

1. **Literal M3 skew.** Raw cubic moment heads recover several signed-tilt constructions; clean score matching finds the direction but attenuates it.
2. **M4 fourth central moment.** Every general route fails at medium information, including direct fourth raw moments. The matched normalized tilt recovers it, demonstrating estimation variance/basis mismatch rather than non-identifiability.

3. **Tail and occupancy.** These are event probabilities, not merely polynomial order. Positivity prevents medium information in the registered family, so their negative result is a detection-boundary result as well as an adapter failure.
4. **G8 path feature.** The effect lives in a structured whole-path motif. An aligned classifier succeeds while generic law samplers and lower-order classifiers miss it. Geometry and representation dominate nominal moment order.

9 Dynamic lag results

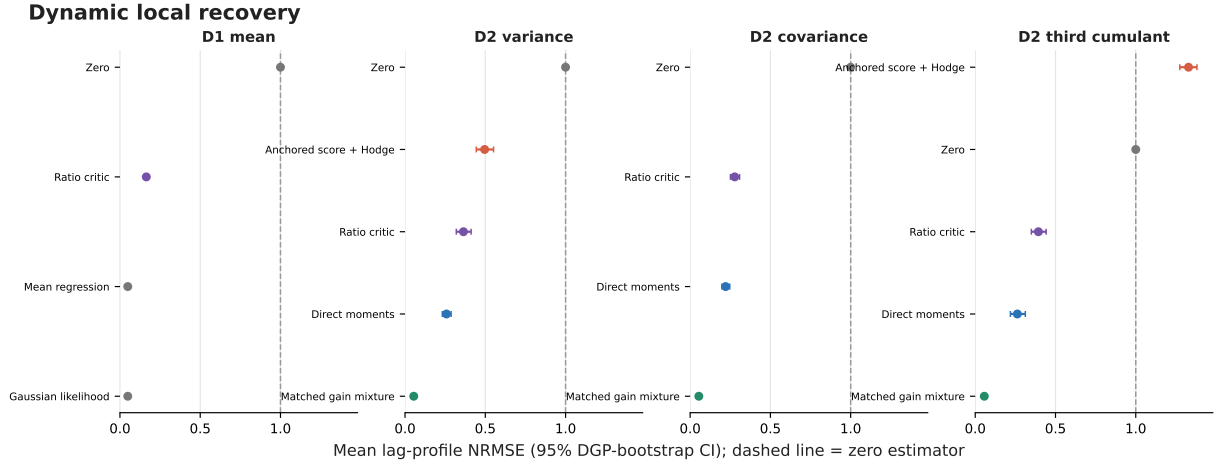


Figure 6: D1/D2 local lag-profile NRMSE with DGP-bootstrap intervals. The matched gain mixture is a specification upper bound; direct moments are the strongest general observational baseline.

D2 source-to-target lag profiles

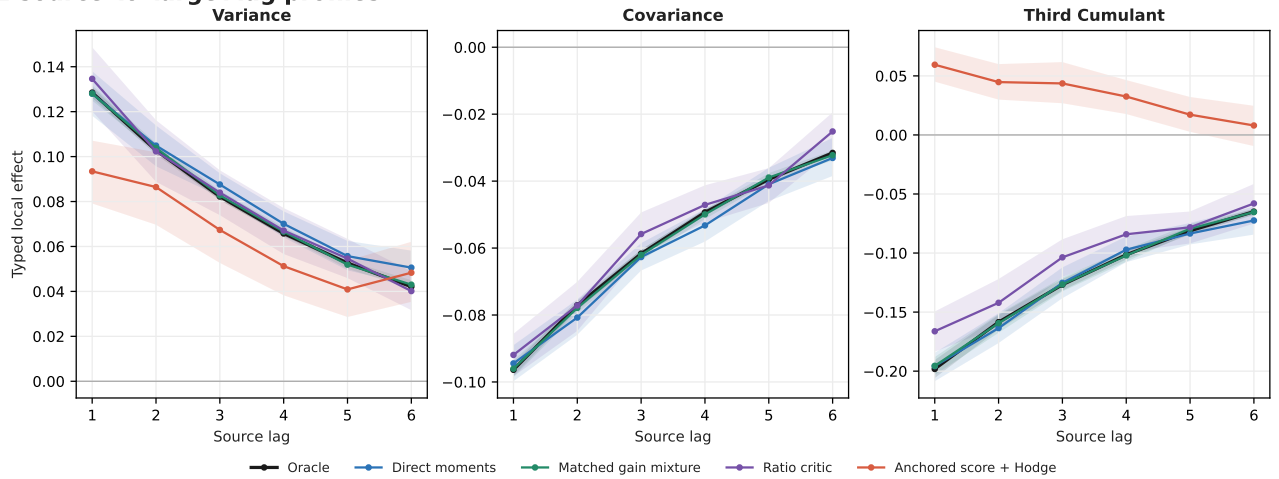


Figure 7: Recovered source-to-target lag profiles, averaged across 30 DGP seeds. The score route follows variance imperfectly and reverses/loses the third-cumulant profile, whereas direct moments and the matched mixture preserve shape and sign.

On the D1 fixed-variance mean anchor, polynomial mean regression and Gaussian likelihood tie at NRMSE about 0.049, while the ratio critic is about 0.164. This is the expected no-advantage result for a correctly specified mean problem.

On D2, the matched mixture achieves NRMSE 0.054–0.057. Direct moments achieve 0.221 for covariance, 0.258 for variance, and 0.262 for third cumulant, all with near-unit slope and sign accuracy above 0.99. The ratio critic is usable but worse. The anchored score has variance NRMSE 0.496 but slope 0.797, just outside the registered interval, and third-cumulant NRMSE 1.329 with a negative slope. Original SBTG field norm is reported only as an unsigned localization diagnostic and is not reinterpreted as a tangent.

10 Baseline comparison and information access

Table 3: Audited information access.

Method class	Effective access	Consequence
Direct moments	observational training	Target-specific raw moment/event heads; analytic derivatives
Ratio critic	training plus conditional evaluation queries	Fresh conditional draws center and integrate the witness
Generic score readouts (Tier 1)	training plus oracle-density quadrature	Favorable best case; not an observational-only comparison
Generic score readouts (Tier 2 static)	training plus oracle-density quadrature	Same favorable quadrature audit as Tier 1
Matched normalized law	known DGP tilt basis	Fits the amplitude of supplied $\psi(y)$
Matched dynamic mixture	known gain/noise family	Fits the mixing law with supplied structural form
Endpoint adapters	endpoint labels or conditional queries	Stronger access than observational local regression

Several nominal method labels in the raw runner understate their access. The audit therefore preserves the recorded label and adds an audited label:

- static and Tier 1 dynamic score readouts use true-DGP density quadrature for centering/integration, making their results favorable best-case score controls;
- ratio readouts use fresh conditional evaluation samples;
- the static matched law knows $\psi(y)$, and the dynamic matched mixture knows gain/noise structure;
- direct moment regression uses observational training and analytic model derivatives.

Consequently no score or matched-model result demonstrates observational advantage over direct moments. The matched-model results instead quantify the benefit of correct structural knowledge.

11 Failure analysis

Table 4: Where each route fails.

Route	First failing stage	Evidence and interpretation
Direct moments	readout variance/basis	M4 quartic NRMSE > 4 ; tail and occupancy fail at low achieved information; works well for M1–M3 and D2
Ratio critic	tangent/readout variance	competitive on simple laws, degrades with context and high-order targets; requires conditional evaluation queries here
Clean Hyvärinen	calibration after Hodge	M3 NRMSE looks favorable but slope is about one half; fourth-order effects fail
Multi-noise DSM	mixed-field derivative	response score is tolerable, history tangent and typed effects are poor
Anchored score	stabilization bias and derivative variance	shrinks M3 toward zero, fails D2 third cumulant, unstable under hidden context, false effect under observation-only null
Endpoint classifier/Riesz	witness variance/representation	generic static adapters fail; aligned G8 classifier succeeds
Matched normalized laws	external validity	excellent in-family, but supplied the mechanism family and therefore not a generic solution

12 Robustness and support

Tier 2 dynamic robustness

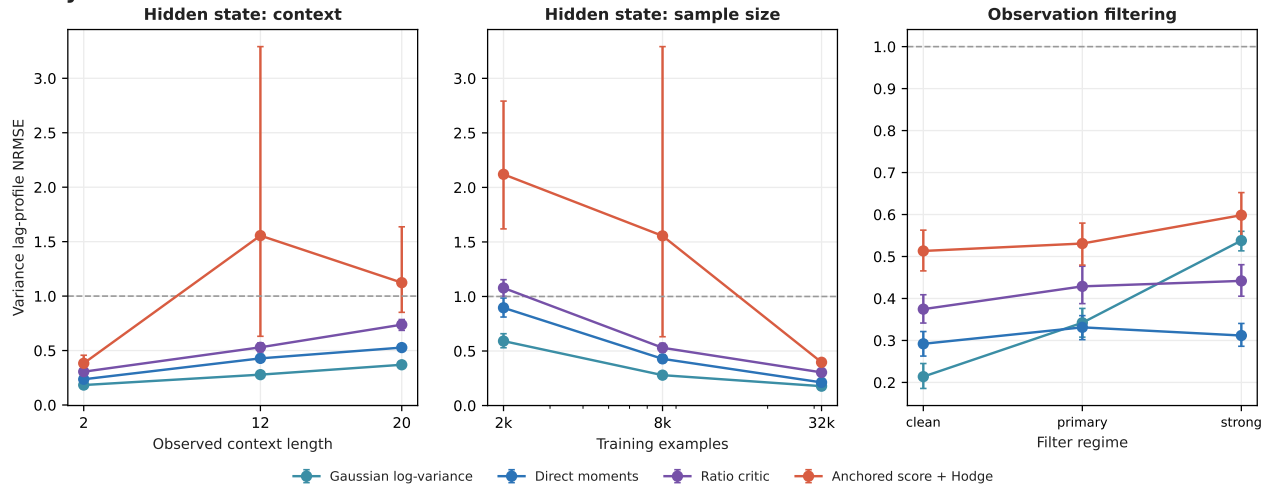


Figure 8: Tier 2 hidden-modulator and observation-filter stresses on new seeds 3001–3030. Direct/log-variance routes scale predictably; the anchored score is sample hungry and sensitive to observation filtering.

Static efficiency, learning, and information slices

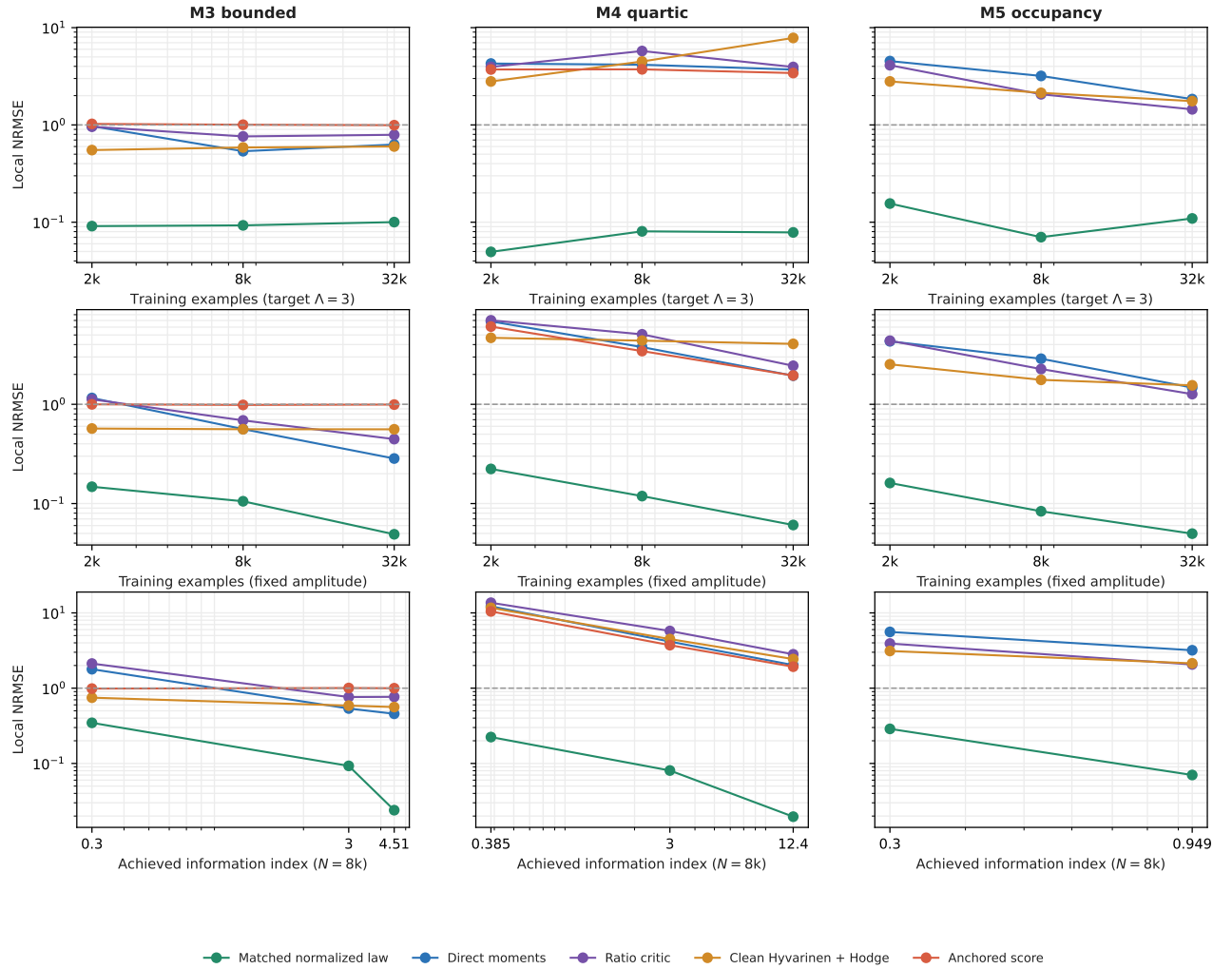


Figure 9: Sample-size and achieved-information slices for bounded M3, M4 quartic, and M5 occupancy. The top row is a preregistered local-alternative sequence: target $\Lambda = 3$ is held fixed by recalibrating amplitude as N changes. The middle row is the post-hoc ordinary learning curve with amplitude fixed at its $N = 8,000$ value. The bottom row varies target information at $N = 8,000$. Positivity bounds make some target values unattainable; actual achieved Λ is used and stored. The design is not a Cartesian product.

Table 5: Selected Tier 2 results: registered rows use seeds 3001–3030; post-hoc fixed-amplitude rows use seeds 4001–4030.

Cell	Method	Metric	Mean [95% CI]	Audited access
D3 context N=8k	2, Gaussian variance	log- nrmse	0.183 [0.152, 0.214]	observational
D3 context N=8k	2, Direct moments	nrmse	0.236 [0.198, 0.272]	observational
D3 context N=8k	2, Anchored score + Hodge	nrmse	0.383 [0.308, 0.457]	observational training plus conditional evalua- tion queries

Cell	Method	Metric	Mean [95% CI]	Audited access
D3 context 12, N=32k	Gaussian log-variance	nrmse	0.178 [0.157, 0.200]	observational
D3 context 12, N=32k	Direct moments	nrmse	0.212 [0.195, 0.229]	observational
D3 context 12, N=32k	Anchored score + Hodge	nrmse	0.397 [0.373, 0.424]	observational training plus conditional evaluation queries
D5 primary filter	Direct moments	nrmse	0.331 [0.302, 0.359]	observational
D5 primary filter	Anchored score + Hodge	nrmse	0.531 [0.480, 0.580]	observational training plus conditional evaluation queries
D5 observation-only null	Anchored score + Hodge	null_rms	0.079 [0.064, 0.102]	observational training plus conditional evaluation queries
M3 bounded, N=2k	Direct moments	NRMSE	0.972 [0.846, 1.108]	observational
M3 bounded, N=32k	Direct moments	NRMSE	0.628 [0.527, 0.732]	observational
M4 quartic, N=32k	Direct moments	NRMSE	3.709 [3.279, 4.138]	observational
M4 quartic, N=32k	Matched normalized law	NRMSE	0.079 [0.058, 0.101]	known dgp tilt basis plus fitted amplitude
M5 occupancy, N=32k	Direct moments	NRMSE	1.843 [1.612, 2.086]	observational
M5 occupancy, N=32k	Matched normalized law	NRMSE	0.109 [0.089, 0.130]	known dgp tilt basis plus fitted amplitude
Post-hoc M3 fixed amp., N=2k	Direct moments	NRMSE	1.158 [0.982, 1.335]	observational
Post-hoc M3 fixed amp., N=32k	Direct moments	NRMSE	0.284 [0.247, 0.323]	observational
Post-hoc M4 fixed amp., N=2k	Direct moments	NRMSE	6.850 [5.594, 8.135]	observational
Post-hoc M4 fixed amp., N=32k	Direct moments	NRMSE	1.943 [1.665, 2.232]	observational
Post-hoc M4 score, N=2k	Clean Hyvärinen + Hodge	NRMSE	4.679 [4.078, 5.413]	observational training plus oracle density quadrature
Post-hoc M4 score, N=32k	Clean Hyvärinen + Hodge	NRMSE	4.066 [3.888, 4.238]	observational training plus oracle density quadrature
Post-hoc M5 fixed amp., N=2k	Direct moments	NRMSE	4.312 [3.756, 4.881]	observational
Post-hoc M5 fixed amp., N=32k	Direct moments	NRMSE	1.474 [1.227, 1.762]	observational

The hidden-modulator experiment tests predictive closure: short context is easier because fewer lag coordinates are requested, sufficient context recovers the full 12-lag source history, and overcomplete context adds eight true null lags. At $N = 32,000$ the anchored score can recover variance numerically,

but it remains less efficient than direct/log-variance estimators. This leaves open whether a more appropriate neural score architecture can close the gap; it does not support the current anchored polynomial implementation as the preferred route.

The static sample-size row is not an ordinary fixed-effect learning curve. It holds total local information near $\Lambda = 3$, so the perturbation amplitude shrinks as N grows. It asks whether each estimator remains efficient under contiguous local alternatives. The separate information row holds $N = 8,000$ and varies target Λ ; positivity bounds prevent event families from reaching every requested regime.

The separately frozen post-hoc fixed-amplitude curve resolves the ambiguity. For M3 bounded skew, direct-moment NRMSE improves from 1.16 at $N = 2,000$ to 0.56 at 8,000 and 0.28 at 32,000, with slope near one throughout; the ratio critic improves from 1.12 to 0.69 to 0.45. Clean score/Hodge stays near 0.56 NRMSE at all three sizes and its slope stays near 0.46–0.48. More data therefore repairs sampling error for the direct/ratio routes but reveals an attenuation plateau in the score route. Because this diagnostic was motivated after the fixed- Λ results were inspected, it is explanatory rather than confirmatory.

M4 quartic is different. At fixed amplitude, direct-moment NRMSE falls from 6.85 to 3.77 to 1.94 as N increases, so sampling error matters, but it still does not beat the zero estimator at 32,000 (achieved $\Lambda \approx 12$). Clean score/Hodge changes only from 4.68 to 4.38 to 4.07. The matched normalized law improves from 0.22 to 0.12 to 0.06. Thus the quartic target is learnable and additional data help the direct estimator, but the registered generic routes are far too inefficient and the score route shows another error plateau.

M5 occupancy improves without crossing the recovery threshold. At fixed amplitude, direct-event NRMSE falls from 4.31 to 2.88 to 1.47, the ratio critic from 4.37 to 2.27 to 1.27, and clean score/Hodge from 2.53 to 1.76 to 1.55. The matched law falls from 0.16 to 0.08 to 0.05. At 32,000, achieved $\Lambda \approx 3.8$ is no longer intrinsically tiny, so the remaining failure is not explained by the original $N = 8,000$ positivity boundary alone.

The observation filter stress is more damaging. Filtering spreads an active variance effect across adjacent source lags, which direct moments still recover. In the matched observation-only null, the score route emits a false variance profile comparable in RMS to the active clean effect. Any real neural-data claim therefore requires an explicit observation-process null and deconvolution or measurement model.

13 Claim registry

Table 6: Preregistered claim registry.

ID	Claim	Result	Evidence
H1	Response-score accuracy need not imply history-tangent accuracy	Yes	Paired operator panels show large error amplification
H2	Direct history tangent can recover typed effects	Partial	Supported for M1–M3 and D1–D3; not M4/tail/occupancy
H3	Stabilized score route has a general advantage	No	No equal-access paired advantage; calibration and null failures
H4	Hodge recovery depends on connected support	Yes	Oracle bridge ladder passes until disconnected endpoint
H5	Finite contrasts can be easier than local derivatives	Partial	True for G8 aligned representation; not generic endpoint adapters
H6	Higher-order success depends on geometry/representation	Yes	M3, M4, events, and path motif separate sharply
H7	Distributional lag recovery is possible	Yes	Matched mixture and direct moments recover D2 profiles

The evidence supports the derivative-gap hypothesis, calibrated direct recovery for several typed targets, topology dependence of Hodge reconstruction, and geometry dependence of higher-order recovery. It rejects a general advantage for the tested stabilized score route and rejects treating original SBTG magnitude as a signed typed tangent. It leaves unresolved generic dense high-dimensional SID, unknown affected subspaces, multi-source/target low-rank factors, and foundation model transfer.

14 Decision

Table 7: Final target-by-target decision table. NRMSE values are mean [95% DGP-bootstrap interval].

Target	Best direct	Best normalized law	Best score-derived	Best finite	Recovered?	Advantage?	Main limitation
Mean	Direct moments 0.52 [0.42, 0.62]	Matched normalized law 0.52 [0.42, 0.62]	Clean Hyvärinen + Hodge 0.32 [0.29, 0.36]	Direct moments (finite) 0.50 [0.41, 0.61]	Numerically yes	No score advantage	Low-order home field; full nuisance FPR not emitted
Variance lag	Direct moments 0.26 [0.23, 0.29]	Matched gain mixture 0.05 [0.05, 0.06]	Anchored score + Hodge 0.50 [0.44, 0.55]	Direct moments (finite) 0.26 [0.23, 0.29]	Yes: direct/matched	Matched only, with stronger structure	Score slope 0.797; hidden state and filtering add error
Covariance lag	Direct moments 0.22 [0.20, 0.25]	Matched gain mixture 0.05 [0.05, 0.06]	No score readout	Direct moments (finite) 0.22 [0.20, 0.25]	Yes: direct/matched	Matched only, with stronger structure	No generic score covariance estimate in the implemented D2 route
Third-cumulant lag	Direct moments 0.26 [0.22, 0.31]	Matched gain mixture 0.06 [0.05, 0.06]	Anchored score + Hodge 1.33 [1.27, 1.38]	Direct moments (finite) 0.26 [0.22, 0.31]	Yes: direct/matched	Matched only, with stronger structure	Score slope is negative; static clean score is attenuated
Standardized skewness	Direct raw moments available	Matched-law moments available	Not separately confirmed	Endpoint moments available	Unresolved as a primary claim	No	Run emitted third cumulant, not a standalone primary skewness board
Tail probability	Direct moments 8.28 [6.96, 9.73]	Matched normalized law 0.19 [0.10, 0.30]	Clean Hyvärinen + Hodge 6.26 [5.63, 6.92]	Direct moments (finite) 8.01 [6.66, 9.39]	Matched law only	No fair general advantage	Positivity capped achieved Λ at 0.128
Mode occupancy	Direct moments 2.57 [2.07, 3.11]	Matched normalized law 0.08 [0.05, 0.12]	Clean Hyvärinen + Hodge 1.85 [1.70, 2.00]	Direct moments (finite) 2.48 [2.01, 2.98]	Matched law only	No fair general advantage	Boundary-limited; achieved Λ at target 3 is 0.949
Fourth central / quartic path	Direct moments 4.44 [3.90, 5.04]	Matched normalized law 0.09 [0.07, 0.11]	Clean Hyvärinen + Hodge 3.97 [3.70, 4.25]	G8 aligned motif MLP: 1.13% median relative error (development)	Matched/access-aligned only	No general advantage	G8 is finite and developmental; it is not the M4 local target
Dynamic lag tensor	Direct moments: 0.22–0.26 across D2 distributional channels	Matched gain mixture: 0.054–0.057	0.50 variance; 1.33 third cumulant	Direct finite: 0.22–0.26	Yes: direct/matched	No score advantage	Dense unknown subspaces and multiple gain factors remain untested

Table 8: Method-selection decision from the implemented evidence.

Use case	Preferred route	Reason
Known mean/covariance/cumulant/event target	Targeted direct head	Best general calibration and efficiency in implemented cells
Many queries from one reusable conditional law	Flexible normalized MDN/flow	Promising next baseline; current matched laws are specification upper bounds
Supported finite intervention with known geometry	Contrast-aligned classifier	G8 development evidence; generic classifiers are insufficient
Operator diagnosis or theory study	Clean score + mixed-field/Hodge gates	Useful decomposition, not current production estimator
Primary general SID estimator	Do not select current DSM/anchored score	No calibrated/equal-access advantage and observation-null failure

Recommended scientific direction. Build the next method around *targeted predictive closure*: a shared history encoder with calibrated heads for means, covariance factors, selected cumulants, tail/occupancy logits, and possibly characteristic-function or kernel features. Benchmark it against a flexible normalized MDN/flow when amortized multi-query use justifies learning the full law. Retain ratio critics for supported contrasts and clean score/Hodge as an operator diagnostic. Do not make the diffusion/DSM score route the primary estimator until it passes mixed-field, calibration, and observation-null gates.

What would change this decision. A convincing score-derived SID result would need: (i) a generic score backend with accurate history mixed fields on at least two independent high-order DGP families; (ii) calibrated typed effects with no oracle density quadrature; (iii) paired advantage over direct and normalized-law baselines under equal access and compute; (iv) robustness to hidden state and observation filters; and (v) dense unknown response geometry rather than a supplied affected coordinate.

15 Limitations and next experiments

1. **Dense D4 benchmark.** Rotate several independent gain factors into an unknown response subspace, use multiple source histories and timescales, and score the full source-by-target-by-lag tensor before low-rank factorization.
2. **Fair flexible laws.** Compare shared-encoder direct heads with a capacity-matched MDN and conditional flow; the current “correct MDN” is an upper bound, not a flexible baseline.
3. **Score architecture.** Replace the compact polynomial score with a time-series network, but gate it first on clean score, mixed-field, and tangent truth. Explore explicit Jacobian regularization, integrability penalties, and joint history-response score formulations.
4. **Readout without oracle quadrature.** Estimate centering and influence functions from held-out observational samples or backend samples, propagate that uncertainty, and use common evaluation budgets.
5. **Finite contrast efficiency.** Use coupled/common-random-number sampling, calibrated density-ratio classifiers, conditional MMD, and analytic feature dictionaries where scientifically justified.
6. **Observation model.** Add calcium/voltage filtering, missingness, heteroskedastic noise, and a matched observation-only null before real-neural-data interpretation.
7. **Mechanism confusion.** Emit every readout on every DGP so the preregistered nuisance false-call requirement is directly estimable rather than inferred from target-only cases.
8. **Unfinished runbook controls.** Complete standalone standardized skewness, the full δ/σ ladders, teacher-contamination and stabilization ablations, heavy-tail/manifold stresses, and multiple stochastic

neural initializations. The compact estimators here are deterministic given a DGP seed and therefore do not substitute for those neural-backend tests.

Further questions

- Is the reusable object really the full response score, or a learned basis of influence functions/conditional characteristic functions that supports many typed queries more directly?
- Can the history tangent be learned directly with normalization/centering constraints, avoiding differentiation of a response score?
- Which neuromodulatory questions require local derivatives, and which are more naturally supported finite interventions?
- How should measurement dynamics and biological slow gain be jointly identified when both create lag spreading?

A Negative results and invalidated executions

Negative results were retained rather than removed from the boards. The principal scientific failures are: all general M4 quartic routes at medium information; all general tail and occupancy routes in their boundary-limited registered families; generic static endpoint classifiers and signed Riesz witnesses; the multi-noise DSM history tangent; anchored-score D2 third cumulants; and the anchored-score observation-only null. No failed confirmatory fit was silently dropped: both Tier 1 and Tier 2 validated with zero failed top-level cases.

Table 9: Developmental or invalidated executions excluded from confirmation.

Run/version	Status	Reason and disposition
Tier 1 v1	Development only	Positive-noise score fields were centered against the wrong law; amendment 1 corrected the declared noised object before confirmation.
Anchored tuning grid	Development only	One global ridge/noise setting was selected on eight development seeds; its poor development performance was retained and the setting frozen.
Tier 1 v3	Invalidated after 9/270 cells	Raw arrays were not persisted. The run was stopped before results were inspected; v4 used identical scientific settings and new immutable CSV/NPZ pairs.
Initial legacy audit launch	Rejected before data	The legacy tier enumeration rejected <code>audit_reproduction</code> ; it was changed to the valid <code>development</code> label without changing scientific settings.
Legacy rerun	Compatibility only	All 117 scientific values were bitwise equal, but source-tree hashes differ and runtime fields are environment-dependent.

The full amendment trail is preserved in `empirical_sid/PREREGISTRATION_AMENDMENT_1.md` through `PREREGISTRATION_AMENDMENT_3.md` and `LEGACY_AUDIT_AMENDMENT_20260714.md`.

B DGP and implementation details

The static primary suite uses response dimension eight, 8,000 training examples, 96 evaluation histories, $\delta = 0.12$, and 30 top-level seeds. M3–M5 effects lie in a registered scalar projection with seven independent nuisance coordinates. Signed tilts are positive normalized laws

$$p_h(y) = \phi(y)\{1 + a \tanh(h)\psi(y)\},$$

with ψ orthogonalized against lower-order nuisance polynomials. This design isolates functional order but does not test discovery of an unknown dense response subspace.

The static direct baseline uses ridge regression on the scalar history basis $1, h, \dots, h^5$ with simultaneous raw-response heads through order four and an additional event-indicator head for tail/occupancy channels. Central-functional derivatives are then computed analytically, including all mean-correction terms. The ratio critic is a joint-versus-shuffled logistic regression using cubic history features, response powers through five, smooth/local event features, and every history–response interaction; its logit derivative is centered with fresh conditional draws. These are deliberately compact estimators, not neural foundation models.

The score models use a tensor-product polynomial basis through cubic history and quintic scaled response, with a noise-coordinate block. Clean Hyvärinen solves the empirical score-matching normal equations. Multi-noise DSM uses corrupted samples at $\sigma \in \{0.03, 0.06, 0.12, 0.25\}$; the anchored version subtracts a Gaussian score constructed from fitted mean/variance heads and regularizes the residual coefficients. The mixed field is a central history secant, and in the one-dimensional affected coordinate Hodge reconstruction is numerical integration followed by mean-zero centering. For the Tier 1 and static Tier 2 score boards, that centering/influence integration uses true-DGP density quadrature, intentionally giving the score route a favorable upper-control readout.

D2 uses a Bernoulli stochastic-gain state centered to preserve conditional mean:

$$Y = a(S - \pi(h)) + \epsilon, \quad \pi(h) = \text{logit}^{-1}(\alpha + k^\top h).$$

Its variance, covariance, and third-cumulant lag derivatives are analytic. D3 integrates a logistic-normal hidden modulator with Gauss–Hermite quadrature. D5 filters independent adjacent latent responses and adds observation noise; its variance derivative is the sum of current and shifted filtered contributions.

C Statistical analysis

All primary intervals resample the top-level DGP seed 2,000 times. Nested query draws are averaged within seed. No confirmatory result selected a hyperparameter; anchored-score ridge and noise schedule were frozen on eight development seeds. Paired comparisons use common DGP seeds. Test outcomes are never pooled with development runs or the aborted pre-raw-array execution.

Table 10: Recorded sequential compute. Wall time is summed once per DGP-family/seed cell; peak memory is the maximum runner-recorded resident usage.

Component	Cells	Total CPU-wall hours	Median sec/cell	Peak MiB
Tier 1 static	270	0.43	7.15	306
Tier 1 dynamic	60	0.01	0.50	n/a
Tier 2 dynamic robustness	270	0.07	0.30	n/a
Tier 2 static slices	450	0.86	6.81	580
Post-hoc fixed-amplitude	270	0.51	6.76	627

All runs were executed sequentially on CPU with one BLAS/OpenMP thread. The Tier 2 static process ran at reduced operating-system priority; completion checks also enforced paired CSV/NPZ counts, SHA-256 agreement, finite metrics, and a minimum 2 GiB free-disk stop condition. The machine-observed process resident set remained below 0.7 GiB.

D Artifact and reproducibility ledger

Table 11: Primary artifact ledger. The displayed hashes are prefixes; complete hashes are in package manifests.

Role	Path	SHA-256 prefix
Tier 1 summary	/Users/vik/Developer/new_sbtg_neuro/empirical_sid/runs/sid_tier1_primary_20260714_v4/results/summary.csv	30dba5e0dcf...
Tier 1 validation	/Users/vik/Developer/new_sbtg_neuro/empirical_sid/runs/sid_tier1_primary_20260714_v4/results/validation.json	f537f7917adf...
Tier 1 package manifest	/Users/vik/Developer/new_sbtg_neuro/empirical_sid/runs/sid_tier1_primary_20260714_v4/results/package_manifest.json	4bae88517c22...
Tier 2 summary	/Users/vik/Developer/new_sbtg_neuro/empirical_sid/runs/sid_tier1_primary_20260714_v4/results/tier2/summary.csv	d0fca3e239a2...
Tier 2 validation	/Users/vik/Developer/new_sbtg_neuro/empirical_sid/runs/sid_tier1_primary_20260714_v4/results/tier2/validation.json	c854efe8cfd...
Tier 2 package manifest	/Users/vik/Developer/new_sbtg_neuro/empirical_sid/runs/sid_tier1_primary_20260714_v4/results/tier2/package_manifest.json	386827fcc276...
Legacy audit	/Users/vik/Developer/new_sbtg_neuro/empirical_sid/runs/sid_tier1_primary_20260714_v4/reports/legacy_audit/legacy_reproduction_audit.json	1c6224b5e886...
Post-hoc diagnostic plan	/Users/vik/Developer/new_sbtg_neuro/empirical_sid/TIER2_POSTHOC_FIXED_AMPLITUDE_PLAN_20260714.md	9f6e0e709b0a...
Machine-readable claims registry	/Users/vik/Developer/new_sbtg_neuro/empirical_sid/runs/sid_tier1_primary_20260714_v4/results/claims_registry.csv	a4ebe4490b3b...

Primary machine-readable outputs are `results/seed_level.parquet`, `results/summary.csv`, `results/tables/paired_advantage.csv`, the Tier 2 package, raw per-seed NPZ archives, manifests, frozen configuration, and source snapshots. SHA-256 inventories and validation JSON files accompany both tiers.

E Runbook completion checklist

Table 12: Runbook completion checklist.

Item	Complete?	Audit note
Frozen preregistration/config/source	Yes	Hashes recorded before confirmatory execution
Stage 0 algebra/software gates	Yes	53 checks passed
Tier 1 static and dynamic cases	Yes	270 static + 30 dynamic; 30 common seeds
Legacy compatibility reproduction	Yes	117 scientific metrics bitwise equal; source hashes differ
Tier 2 hidden-state/observation stresses	Yes	270 dynamic + 450 static registered cases
Post-hoc fixed-amplitude diagnostic	Yes	270 exploratory cases on new seeds; cannot upgrade confirmation
DGP-seed bootstrap and access audit	Yes	2,000 replicates; effective access retained
Full learned mechanism-confusion matrix	No	Target-only cells cannot certify nuisance FPR
Dense unknown response geometry (D4)	No	Affected coordinate is supplied in current static suite
Flexible MDN/flow and neural score backends	No	Still proposed work
Full stabilization/teacher-contamination ablations	No	Frozen compact score shortlist only
Standalone skewness and full delta/noise ladders	No	Primary registered cells only

The program is complete for the implemented Tier 1 and registered Tier 2 cells. The broad runbook completion claim remains withheld because full learned mechanism-confusion matrices, dense D4 geometry, flexible MDN/flow robustness, and general neural score backends were not completed. The scientifically correct label is therefore **diagnostic empirical validation**, not universal moment-recovery validation.